Memo: Regex model vs Gemini model for used-car pricing

To: Pricing team · **From:** Dave Wanik · **Date:** October 14, 2026
Data: 467 Hartford-area Craigslist listings, Sep 28 to Oct 14 (461 after removing junk prices and placeholder mileage)

Bottom line

The model built on Gemini's features is **about $150 more accurate per car** than the regex model, and the result is not luck: it won **21 of 25** paired cross-validation folds (95% interval for the gain: **$66 to $234**). **Recommend shipping the Gemini model.** The extra cost is about $2 per 1,000 listings.

Results

Test	Regex model (4 fields)	Gemini model (14 fields)	Difference
Newest cars (Oct 8 to 14, 87 cars), MAE	$2,394	$2,246	Gemini better by $148 (6%)
25 paired folds, all cars, average MAE	$3,074	$2,923	Gemini better by $150, wins 21 of 25
Older cars only (training window), CV MAE	$3,217	$3,253	Tie

Both champions are random forests, tuned with the same grids, seed and data. Guessing the median price misses by $3,812.

Why the Gemini model wins

- **All of the gain comes from the 10 new fields, not from better reading.** Gemini's version of the same four fields scored *worse* than regex ($3,174 vs $3,074): its longer model names ("f-150 supercrew", "ram 1500") split cars into rarer groups.
- **What the model uses:** mileage, year and make still lead. Among the new fields, **drivetrain, body type, city and condition** matter. Transmission, leather seats and title status add almost nothing yet.
- **The patterns make sense:** with everything else held the same, trucks price about $1,250 above sedans, and front-wheel drive about $1,300 below four-wheel and rear-wheel drive.

Would more data help?

Yes, for both models. Retrained on 92 to 368 cars, both error curves are still falling. The Gemini model is ahead at every size, by roughly $170 to $250 once there are 200 or more cars. More data mainly helps rare categories: with only 15 rebuilt titles, the model cannot yet learn the discount a rebuilt title should carry.

Cost

	Regex	Gemini
Per 1,000 listings	essentially $0	about $2
Time per listing	milliseconds	about 5 seconds
One-time re-read of all 467 old listings	none	about $1, 16 minutes

Recommendation and next steps

1. **Ship the Gemini model** to production (`predict-shap` reads its features from the model card; no code changes).
2. **Re-run this comparison at about 1,000 listings** to confirm the gap holds as both models improve.
3. **Clean Gemini's model names** (for example, "f-150 supercrew" to "f-150") to recover the reading loss above.
4. **Watch rebuilt and salvage titles** as they accumulate; that discount should become learnable.